

Lec: 07

Applications of Matrices

matrix inverse

Solve system of linear equations

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n$$

$x_1, x_2, \dots, x_n$
n variables
n unknowns.

Linear System: Linear in
 $x_1, x_2, \dots, x_n$

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}_{\vec{x}} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}}_{\vec{b}}$$

$$\boxed{A \vec{x} = \vec{b}}$$

$n \times n$

if A^{-1} exists
Solution $\bar{x} = A^{-1}\bar{b}$

Example:

$$\begin{bmatrix} 7 & 2 & 1 \\ 0 & 3 & -1 \\ -3 & 4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

$$\begin{aligned} 7x_1 + 2x_2 + x_3 &= 2 \\ 3x_2 - x_3 &= -1 \\ -3x_1 + 4x_2 - 2x_3 &= 3 \end{aligned}$$

Solution : $A^{-1}\bar{b}$

$$= \begin{bmatrix} -2 & 8 & -5 \\ 3 & -11 & 7 \\ 9 & -34 & 21 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

(A^{-1}) $\underbrace{\quad}_{\bar{b}}$

$$\bar{x} = \begin{bmatrix} -27 \\ 38 \\ 115 \end{bmatrix}$$

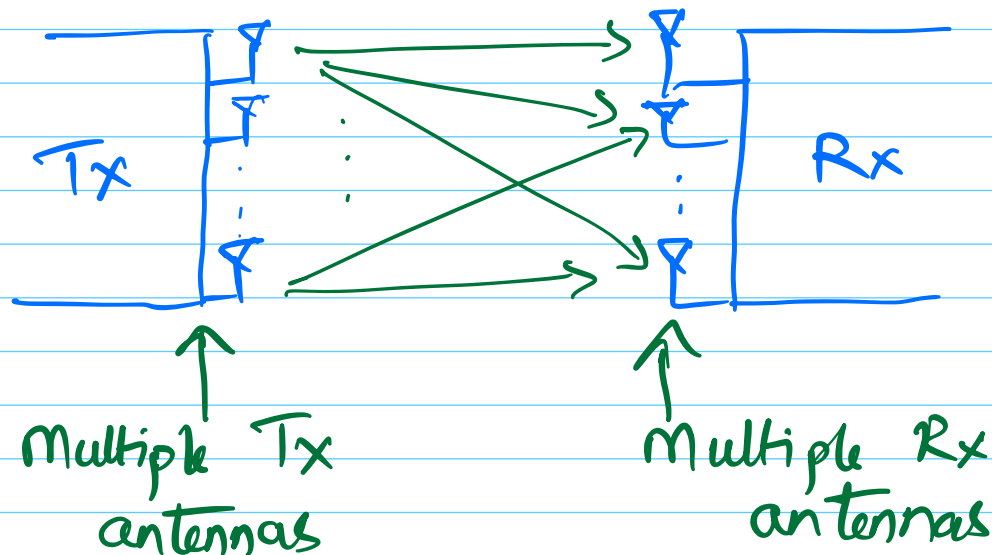
PRACTICAL APPLICATIONS

Several applications : MATRICES

Ex: Wireless Systems

Modern Wireless networks
4G / 5G
→ mimo technology

mimo: Multiple input
Multiple output



mimo: Very important tech
4G / 5G

↑ makes possible to
transmit very high
data rates.

h_{ij} = channel coeff.
b/w Rx antenna i &
Tx antenna j

h_{23} = Channel coeff. b/w
Rx ant. 2 & Tx ant. 3

$$H = \begin{bmatrix} h_{11} & h_{12} & \dots & h_{1t} \\ h_{21} & h_{22} & \dots & h_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ h_{r1} & h_{r2} & \dots & h_{rt} \end{bmatrix}$$

$r \times t$

$r = \#$ Receive antennas

$t = \#$ Transmit antennas

$$\text{rank}(H) \leq \min(r, t).$$

maximum number
of symbols
Transmitted $\leq \text{rank}(H)$.

Determines the data
rate

large $\text{rank}(H) \Rightarrow$ Large
Data rate

MIMO system $\Rightarrow$ Transmit or multiplex
large no. of symbols
in same time/freq.

This property of MIMO is
termed as

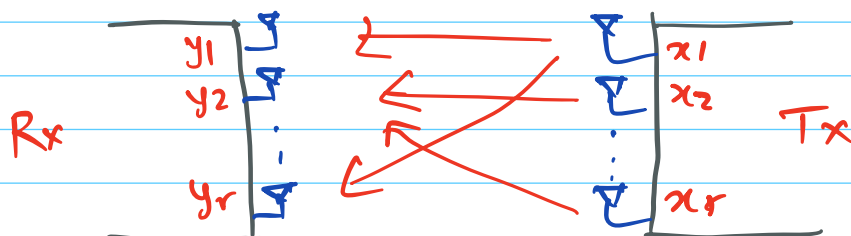
SPATIAL multiplexing

$\rightarrow$ makes it possible to transmit
at very high data rates in

4G/5G.

Parallel transmission of multiple streams.

MIMO Receiver



$r = \text{no. of Tx. ant.}$
 $= \text{no. of Rx. ant.}$

$x_1, x_2, \dots, x_r$
 Transmitted symbols

$y_1, y_2, \dots, y_r$
 Received symbols

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_r \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & \dots & h_{1r} \\ h_{21} & h_{22} & \dots & h_{2r} \\ \vdots & \vdots & \ddots & \vdots \\ h_{r1} & h_{r2} & \dots & h_{rr} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_r \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \\ \vdots \\ n_r \end{bmatrix}$$

$\bar{y} \qquad \qquad H \qquad \qquad \bar{x} \qquad \qquad \bar{n}$

$$\bar{y} = H \bar{x} + \bar{n}$$

MIMO system model

$\bar{y}$ (output vector) = H (channel matrix) $\bar{x}$ (Transmit vector) + $\bar{n}$ (noise vector)

$r \times r$ (special case)

How to determine $\bar{x}$ at Receiver
given $\bar{y}$?

if H is invertible

$$\hat{x} = H^{-1} \bar{y}$$

what if $r > t$?

Things become
interesting!

(we will see this
in future modules)

valid only when
 $r = t$

H is invertible

ZF (zero forcing) receiver